FLATNESS MODULO AN IDEAL PLUS VANISHING OF TOR ONE IMPLIES FLATNESS ON NILPOTENTLY ANNIHILATED MODULES

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Allowed standard prerequisites (editorial).

- Flatness tested on quotients by powers of the ideal. Used in the first part of the counted proof. Reduces flatness of the module modulo each power to a Tor computation.
- Long exact sequences of Tor for a short exact sequence. Used throughout the counted proof. Transports vanishing of Tor one from the quotient by the ideal to the powers and to modules annihilated by a power.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal. Let M be an R-module. If M/IM is flat over R/I and $Tor_1^R(R/I, M) = 0$ then*

(1) *M/I^nM is flat over R/I^n for all $n \geq 1$, and*
(2) *for any module N which is annihilated by I^m for some $m \geq 0$ we have $Tor_1^R(N, M) = 0$.*

In particular, if I is nilpotent, then M is flat over R.

Proof. Assume M/IM is flat over R/I and $\mathrm{Tor}_1^R(R/I, M) = 0$. Let N be an R/I-module. Choose a short exact sequence

$$0 \to K \to \bigoplus\nolimits_{i \in I} R/I \to N \to 0$$

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By the long exact sequence of Tor and the vanishing of $\mathrm{Tor}_1^R(R/I, M)$ we get

$$0 \to \mathrm{Tor}_1^R(N, M) \to K \otimes_R M \to (\bigoplus\nolimits_{i \in I} R/I) \otimes_R M \to N \otimes_R M \to 0$$

But since K, $\bigoplus_{i \in I} R/I$, and N are all annihilated by I we see that

$$K \otimes_R M = K \otimes_{R/I} M/IM,$$
$$(\bigoplus\nolimits_{i \in I} R/I) \otimes_R M = (\bigoplus\nolimits_{i \in I} R/I) \otimes_{R/I} M/IM,$$
$$N \otimes_R M = N \otimes_{R/I} M/IM.$$

As M/IM is flat over R/I we conclude that

$$0 \to K \otimes_{R/I} M/IM \to (\bigoplus\nolimits_{i \in I} R/I) \otimes_{R/I} M/IM \to N \otimes_{R/} M/IM \to 0$$

is exact. Combining this with the above we conclude that $\mathrm{Tor}_1^R(N, M) = 0$ for any R-module N annihilated by I.

Let us prove (2) by induction on m. The case $m = 1$ was done in the previous paragraph. For N annihilated by I^m for $m > 1$ we may choose an exact sequence $0 \to N' \to N \to N'' \to 0$ with N' and N'' annihilated by I^{m-1}. For example one can take $N' = IN$ and $N'' = N/IN$. Then the exact sequence

$$\mathrm{Tor}_1^R(N', M) \to \mathrm{Tor}_1^R(N, M) \to \mathrm{Tor}_1^R(N'', M)$$

and induction prove the vanishing we want.

Finally, we prove (1). Given $n \geq 1$ we have to show that M/I^nM is flat over R/I^n. In other words, we have to show that the functor $N \mapsto N \otimes_{R/I^n} M/I^nM$ is exact on the category of R-modules N annihilated by I^n. However, for such N we have $N \otimes_{R/I^n} M/I^nM = N \otimes_R M$. By the vanishing of Tor_1 in (2) we see that the functor $N \mapsto N \otimes_R M$ is exact on the category of N annihilated by some power of I and we conclude. $\square$